1 RQ1a: Corruption Detection - Citation

1.1 Comprehensive Analysis

This comprehensive 6-panel visualization presents all key metrics for citation-based corruption detection: F1 scores, AUC-ROC, precision-recall trade-off, detection recall by corruption type, score discrimination, and pairwise prompt comparisons with effect sizes.

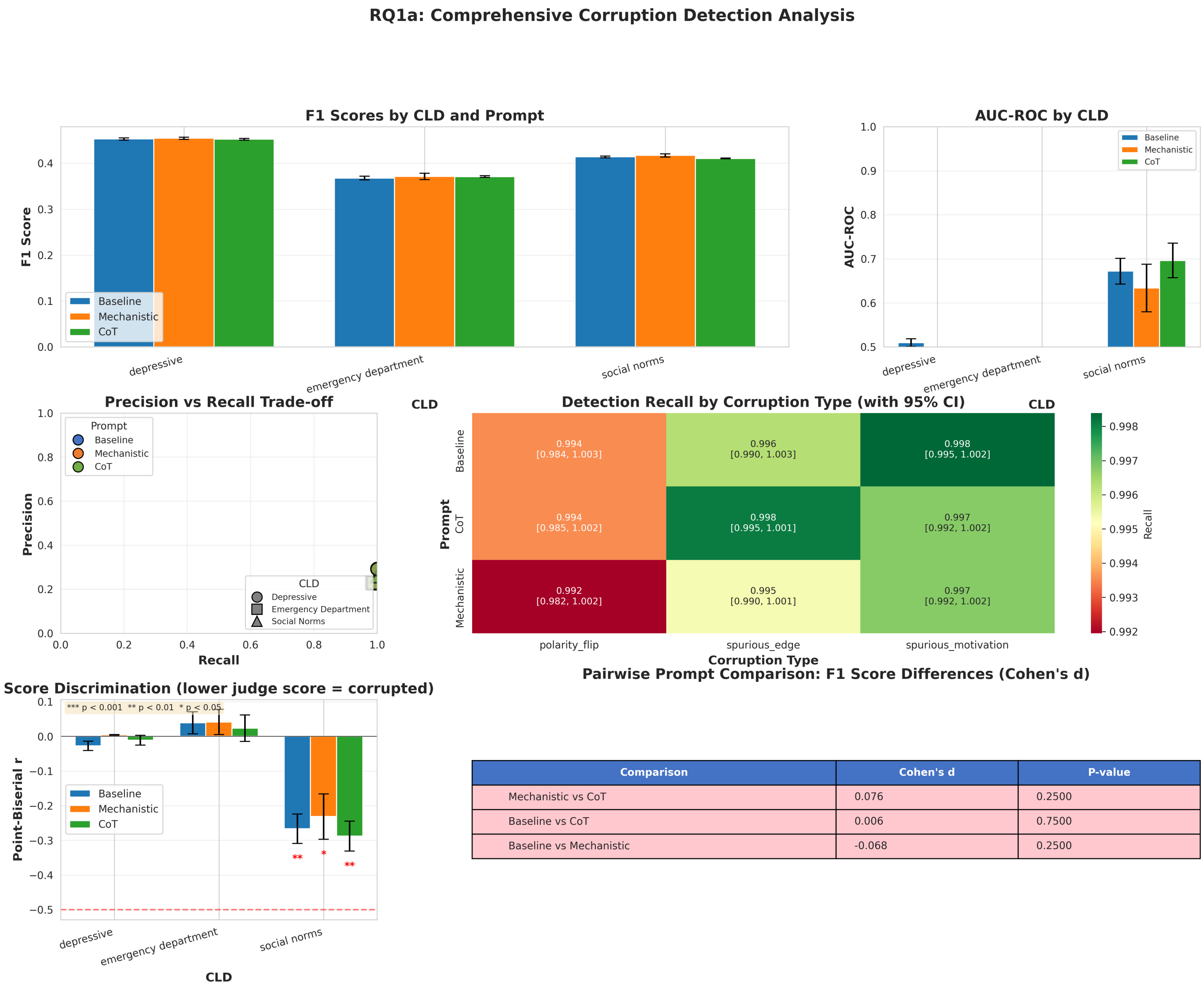


Figure 1: Comprehensive corruption detection analysis for citation-based validation. Six panels show: (top left) F1 scores by CLD and prompt, (top center) AUC-ROC scores, (top right) precision-recall trade-off, (middle left) detection recall by corruption type with 95% CI, (middle right) score discrimination via point-biserial correlations, (bottom) pairwise prompt comparisons with Cohen's d effect sizes.

1.2 ROC Curves

Detailed ROC curves showing true positive rate vs false positive rate for each prompt variant across all CLDs.

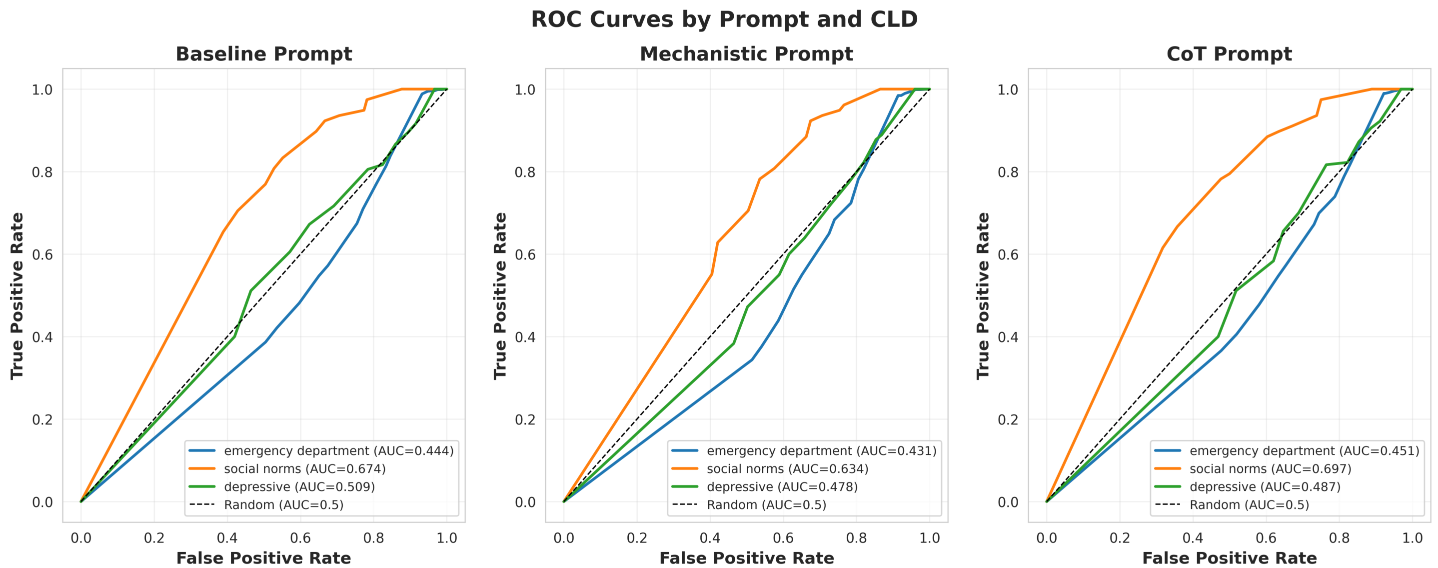
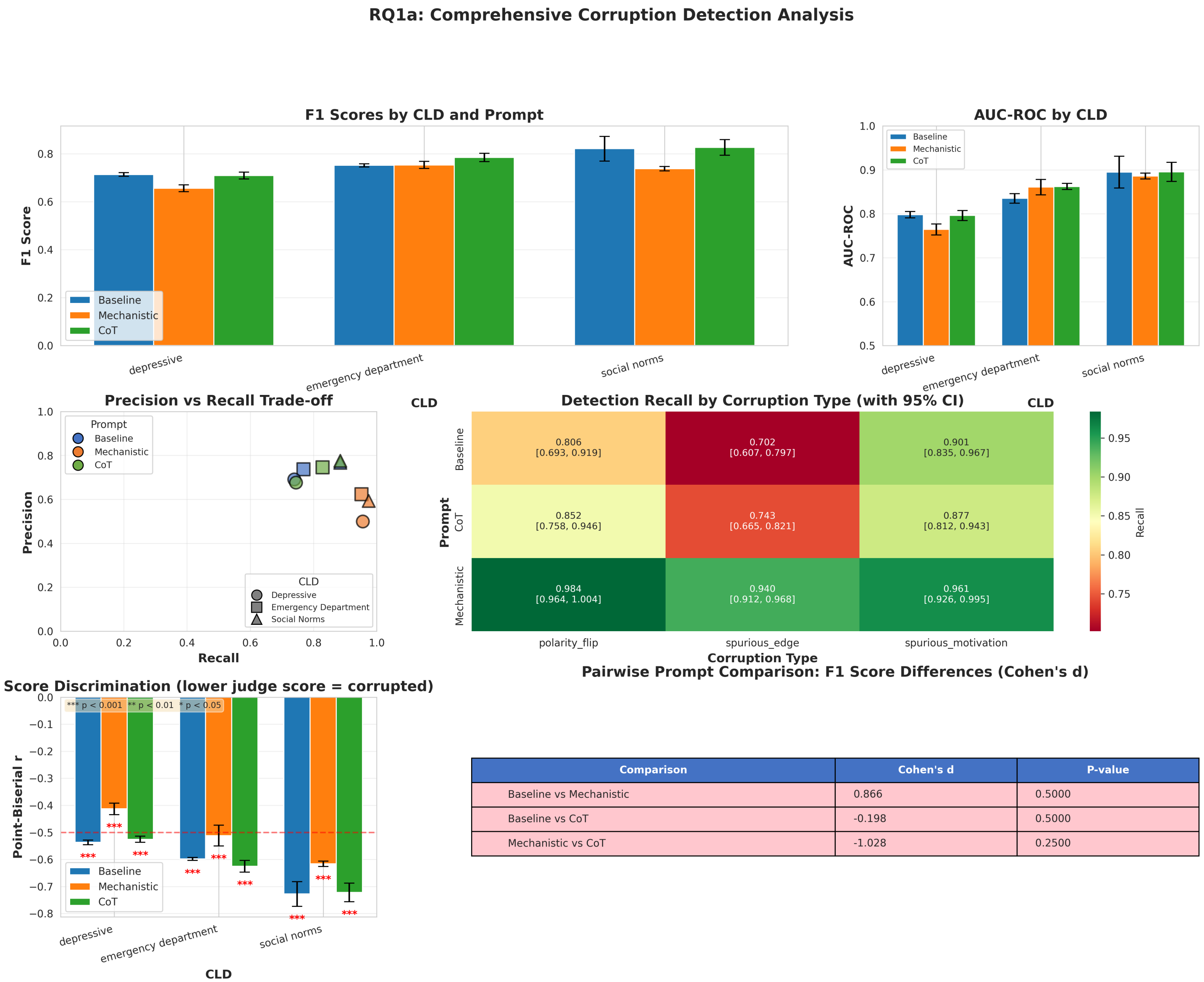


Figure 2: ROC curves for citation-based corruption detection by prompt. Each subplot shows performance for one prompt variant (Baseline, Mechanistic, CoT) across all three CLDs. Curves above the diagonal (chance line) indicate better-than-random discrimination.

2 RQ1a: Corruption Detection - Correctness

2.1 Comprehensive Analysis

This comprehensive 6-panel visualization presents all key metrics for correctness-based corruption detection.



Comparison	Cohen's d	P-value
Baseline vs Mechanistic	0.866	0.5000
Baseline vs CoT	-0.198	0.5000
Mechanistic vs CoT	-1.028	0.2500

Figure 3: Comprehensive corruption detection analysis for correctness-based validation. Six panels show: (top left) F1 scores by CLD and prompt, (top center) AUC-ROC scores, (top right) precision-recall trade-off, (middle left) detection recall by corruption type with 95% CI, (middle right) score discrimination via point-biserial correlations, (bottom) pairwise prompt comparisons with Cohen's d effect sizes.

2.2 ROC Curves

Detailed ROC curves showing discrimination performance for each prompt variant.

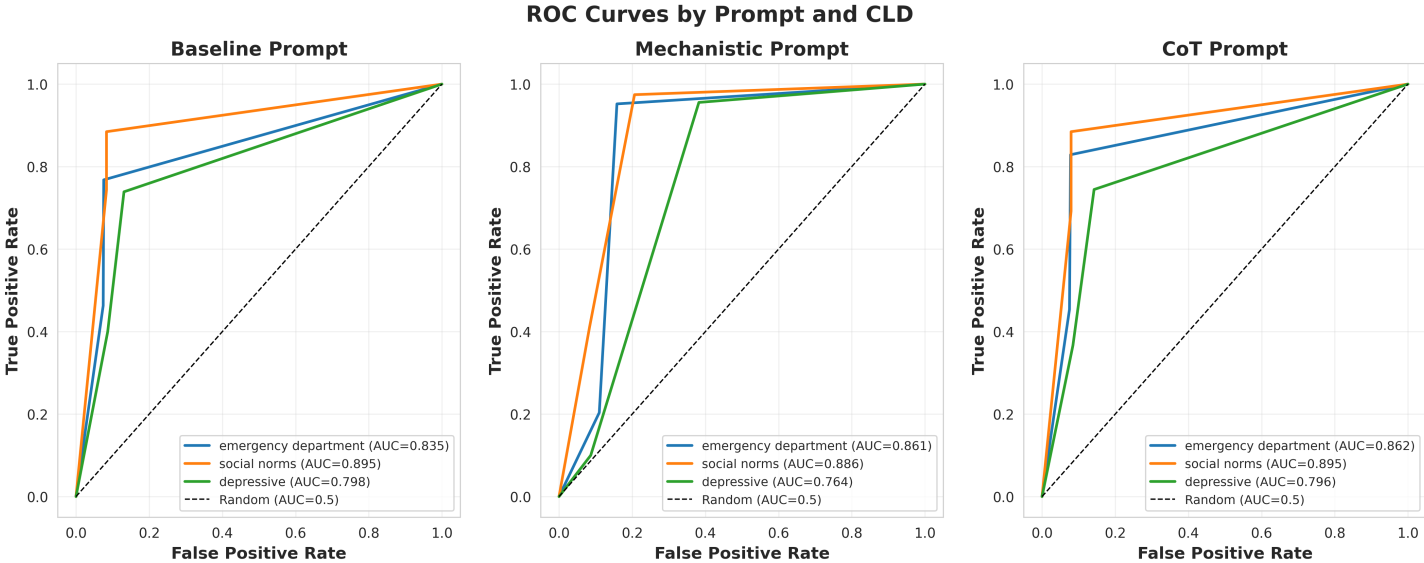


Figure 4: ROC curves for correctness-based corruption detection by prompt. Each subplot shows performance for one prompt variant across all CLDs.

3 RQ1a: Ground Truth - Citation

3.1 Comprehensive Analysis

Ground truth experiments establish baseline performance without artificial corruption. This 6-panel visualization includes all prompt variants tested (baseline, cot, mechanistic_lit, mechanistic_original).

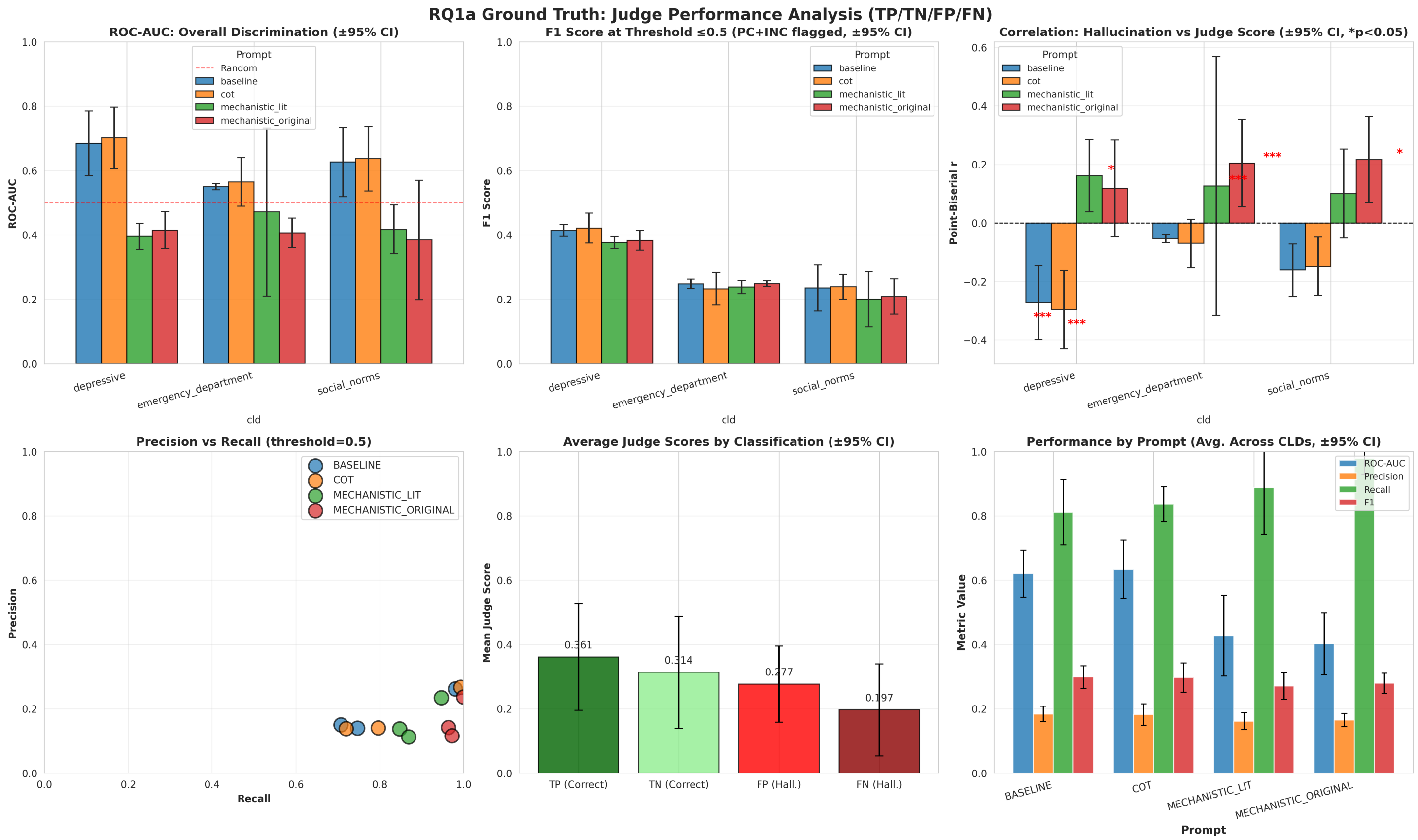


Figure 5: Comprehensive ground truth analysis for citation validation. Six panels show: (top row) ROC-AUC, F1 scores, and point-biserial correlations by CLD and prompt; (bottom row) precision-recall trade-off, score distributions by classification type, and performance by prompt across CLDs. All metrics include 95% confidence intervals.

3.2 Score Heatmap

Detailed heatmap showing judge score patterns by classification type across all CLD and prompt combinations.

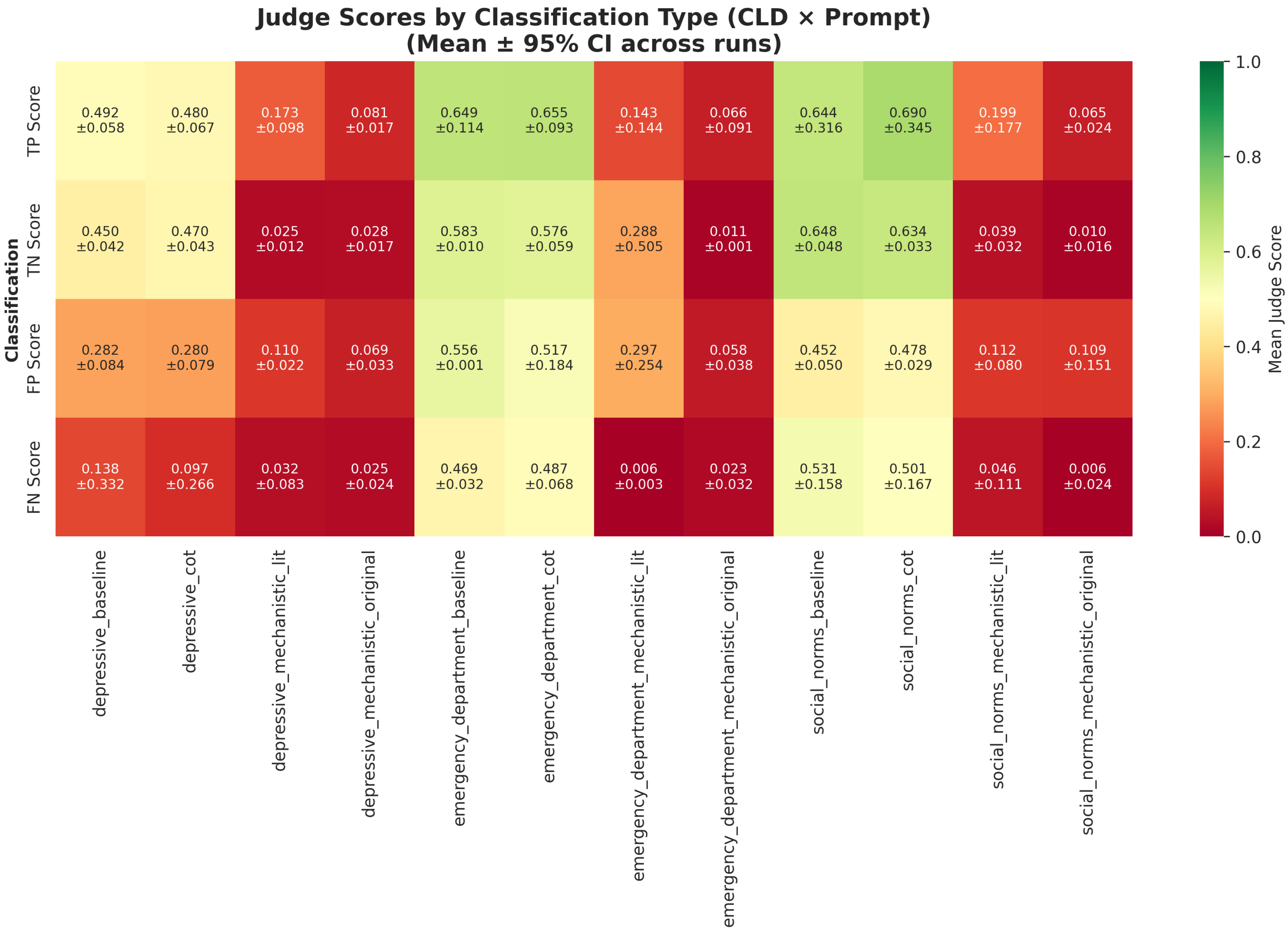


Figure 6: Score heatmap for ground truth citation validation. Shows mean judge scores with 95% confidence intervals for each classification type (TP, TN, FP, FN) across all CLD × prompt combinations.

4 RQ1a: Ground Truth - Correctness

4.1 Comprehensive Analysis

Ground truth experiments with correctness-based validation, establishing baseline detection performance.

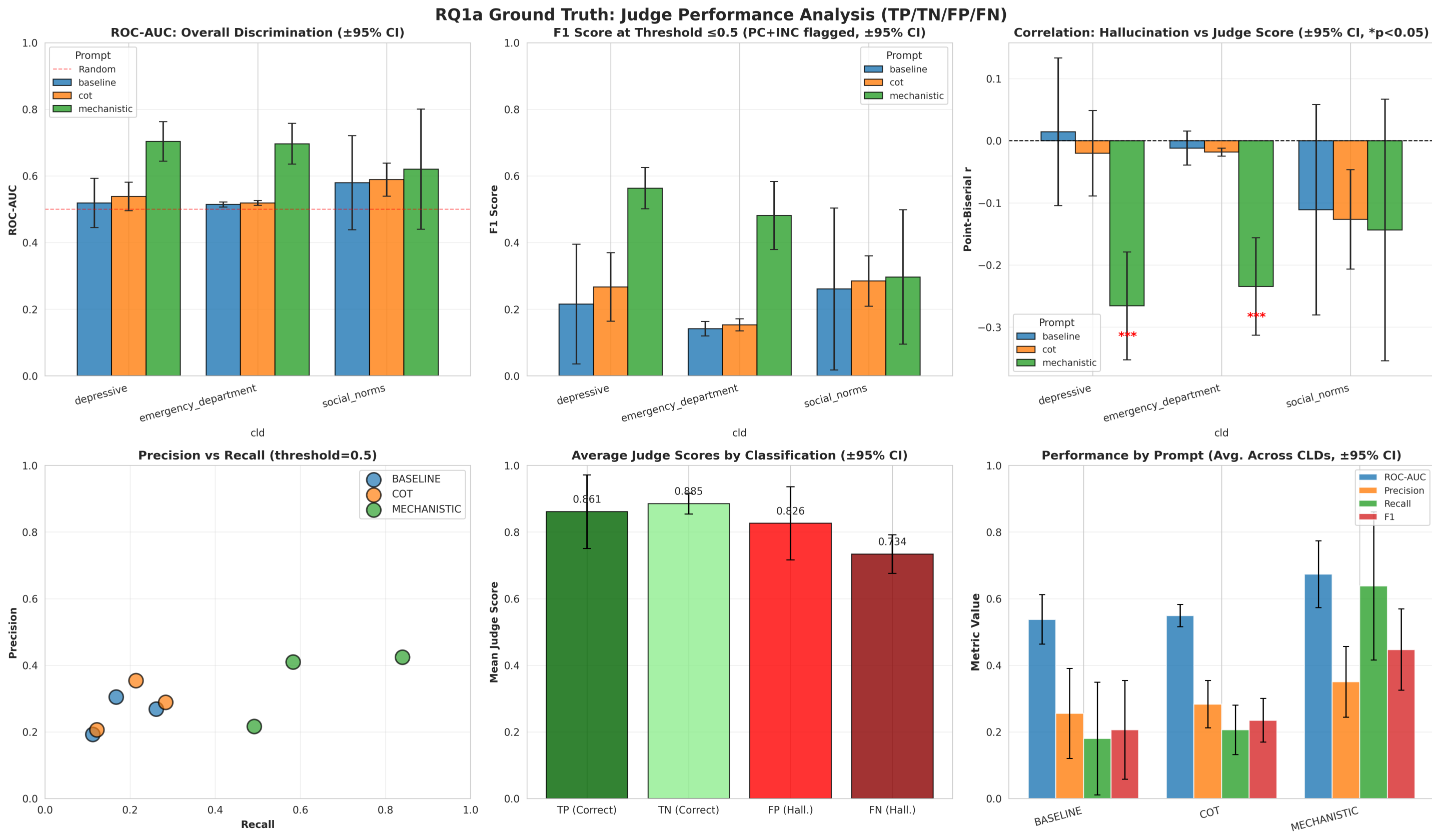


Figure 7: Comprehensive ground truth analysis for correctness validation. Six panels show: (top row) ROC-AUC, F1 scores, and point-biserial correlations; (bottom row) precision-recall trade-off, score distributions, and aggregate performance metrics. All with 95% confidence intervals.

4.2 Score Heatmap

Detailed score patterns across all experimental conditions.

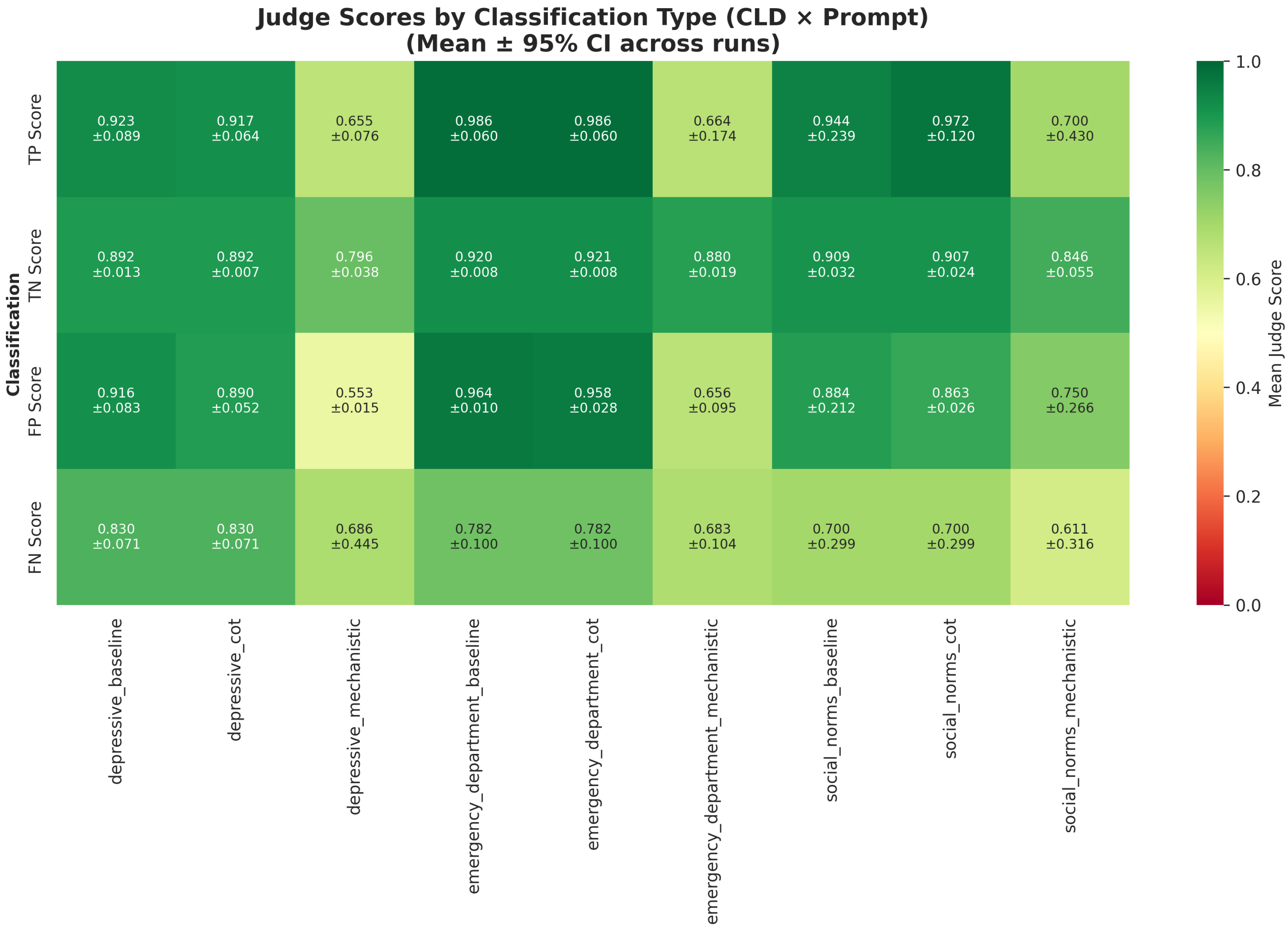


Figure 8: Score heatmap for ground truth correctness validation showing classification scores across CLD × prompt combinations.

5 RQ1b: Corrector Experiments

5.1 Ground Truth Correctness Baseline Results

Results table showing corrector performance on ground truth data using correctness-based evaluation.

RQ1b Corrector Results: LLM-Generated CLDs vs Excel Ground Truth
(Correctness Judging, Baseline Prompt, 3 runs per CLD, 95% CI)

	F1 Δ	Precision Δ	Recall Δ	Judge Score Δ	Total Actions	Revise	Change Type
Social Norms	-0.044 ± 0.087	-0.114 ± 0.079	+0.083 ± 0.094	+0.083 ± 0.024	12.0 ± 3.4	2.3 ± 0.7	8.7 ± 3.5
Depressive	-0.036 ± 0.005	-0.043 ± 0.005	+0.029 ± 0.000	+0.100 ± 0.005	27.0 ± 2.0	14.7 ± 1.3	12.3 ± 1.7
Emergency Dept	+0.030 ± 0.008	+0.010 ± 0.004	+0.131 ± 0.026	+0.079 ± 0.005	100.3 ± 6.6	23.7 ± 4.6	76.0 ± 7.9

Figure 9: Results table for RQ1b corrector experiment on ground truth data with correctness-based validation. Shows correction accuracy and performance metrics.

5.2 Synthetic Corruption Correctness Results

Results table showing corrector performance on synthetically corrupted data.

RQ1b: Corrector on Synthetic Corrupted CLDs (Baseline Prompt)

CLD	F1 Δ	Precision Δ	Recall Δ	Judge Score Δ	Total Actions	Revise	Change Type
Social Norms	+0.000 ± 0.000	+0.000 ± 0.000	+0.000 ± 0.000	+0.028 ± 0.108	4.3 ± 8.0	3.3 ± 3.8	1.0 ± 4.3
Depressive	+0.145 ± 0.033	+0.164 ± 0.120	+0.117 ± 0.112	+0.186 ± 0.030	63.0 ± 13.8	31.7 ± 18.3	30.7 ± 3.8
Emergency Dept	+0.082 ± 0.045	+0.062 ± 0.031	+0.121 ± 0.085	+0.135 ± 0.010	268.0 ± 26.2	115.0 ± 17.9	152.7 ± 16.0

Figure 10: Results table for RQ1b corrector experiment on synthetically corrupted data. Evaluates the system's ability to recover from known corruptions.

6 Sensitivity Analysis

6.1 Aggregate Sensitivity Comparison

Aggregate comparison of sensitivity across all experimental conditions.

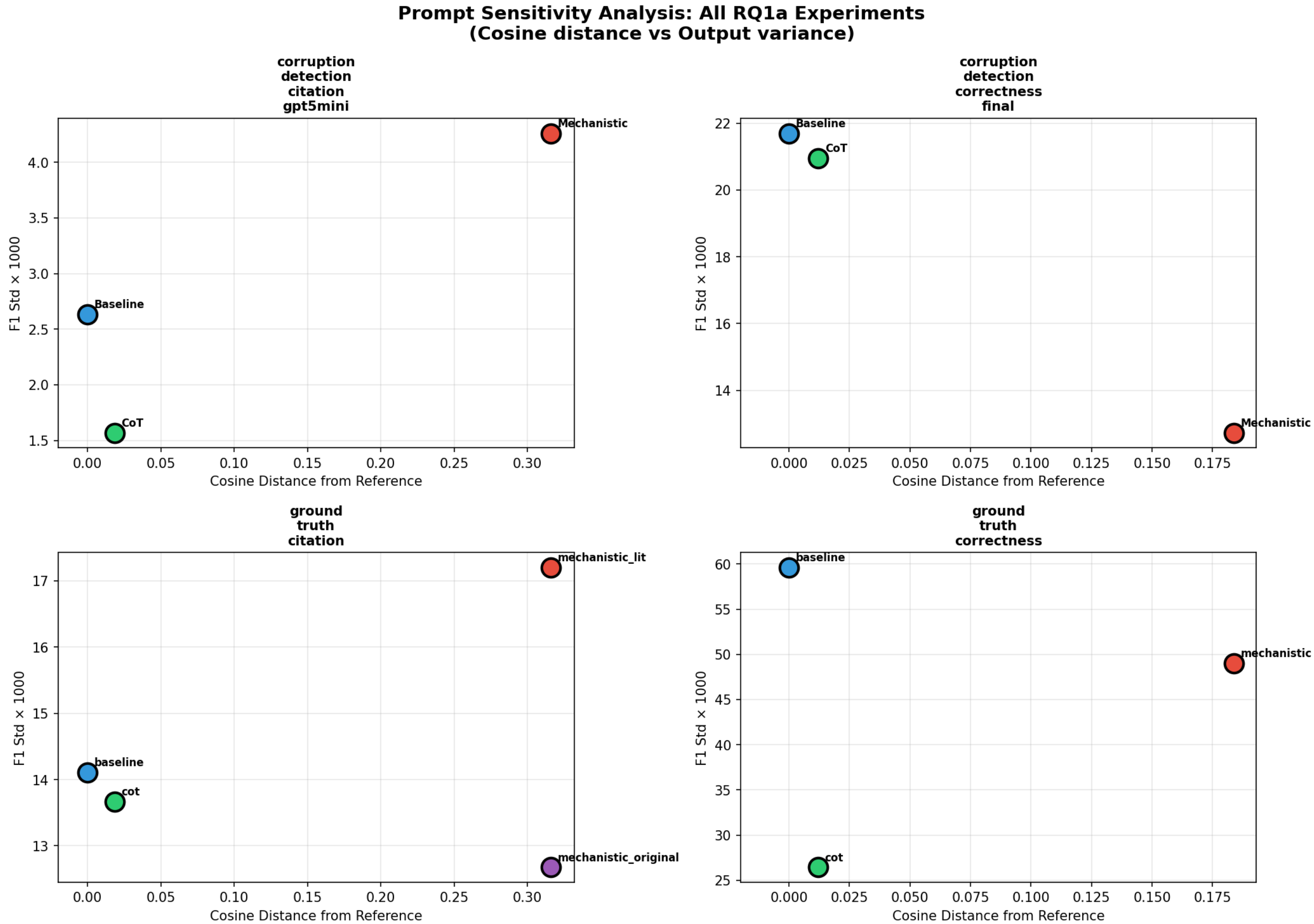


Figure 11: Aggregate sensitivity analysis comparing parameter robustness across all experiments. Shows which parameters have the greatest impact on system performance.

6.2 Citation Detection Sensitivity

Sensitivity analysis specific to citation-based corruption detection.

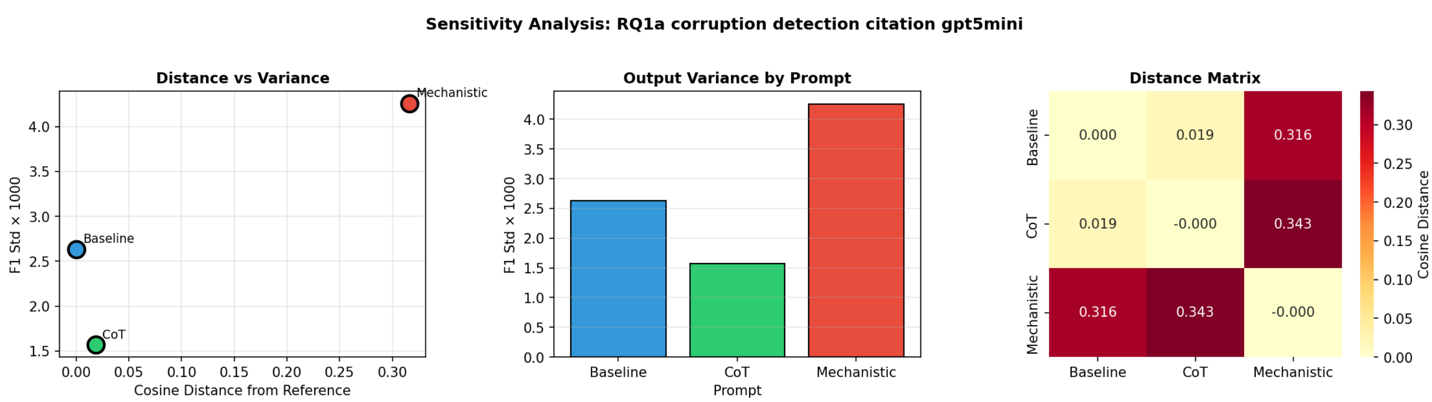


Figure 12: Sensitivity analysis for citation-based corruption detection. Evaluates parameter stability for the citation validation approach.

6.3 Correctness Detection Sensitivity

Sensitivity analysis specific to correctness-based corruption detection.

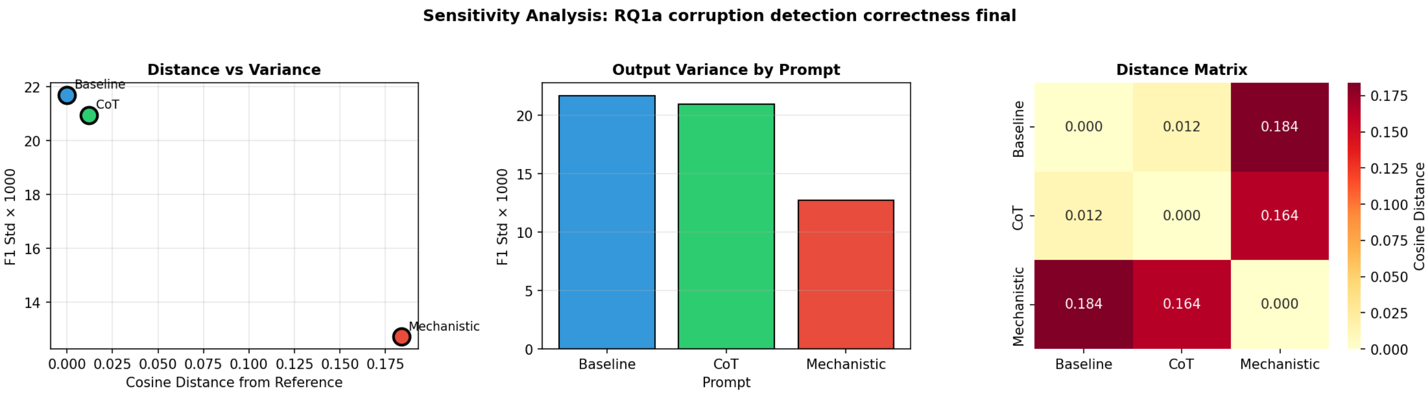


Figure 13: Sensitivity analysis for correctness-based corruption detection. Shows parameter robustness for the correctness validation approach.

6.4 Ground Truth Citation Sensitivity

Sensitivity analysis for ground truth experiments with citation validation.

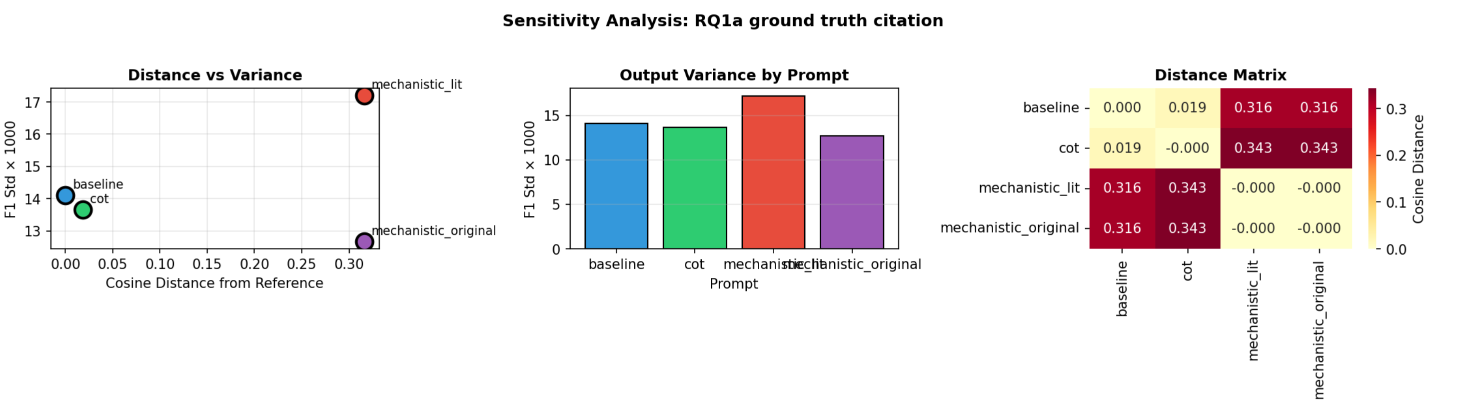


Figure 14: Sensitivity analysis for ground truth citation validation experiments. Assesses baseline parameter stability.

6.5 Ground Truth Correctness Sensitivity

Sensitivity analysis for ground truth experiments with correctness validation.

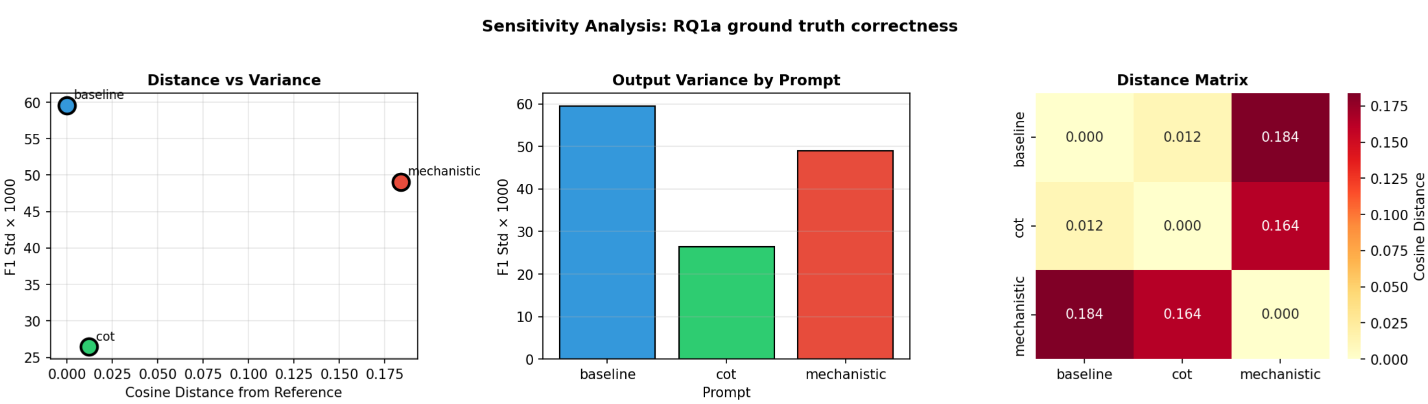


Figure 15: Sensitivity analysis for ground truth correctness validation experiments. Evaluates baseline parameter robustness.

7 Sobol Sensitivity Analysis

7.1 Aggregate Sobol Comparison

Aggregate Sobol sensitivity analysis comparing variance contributions across experiments.

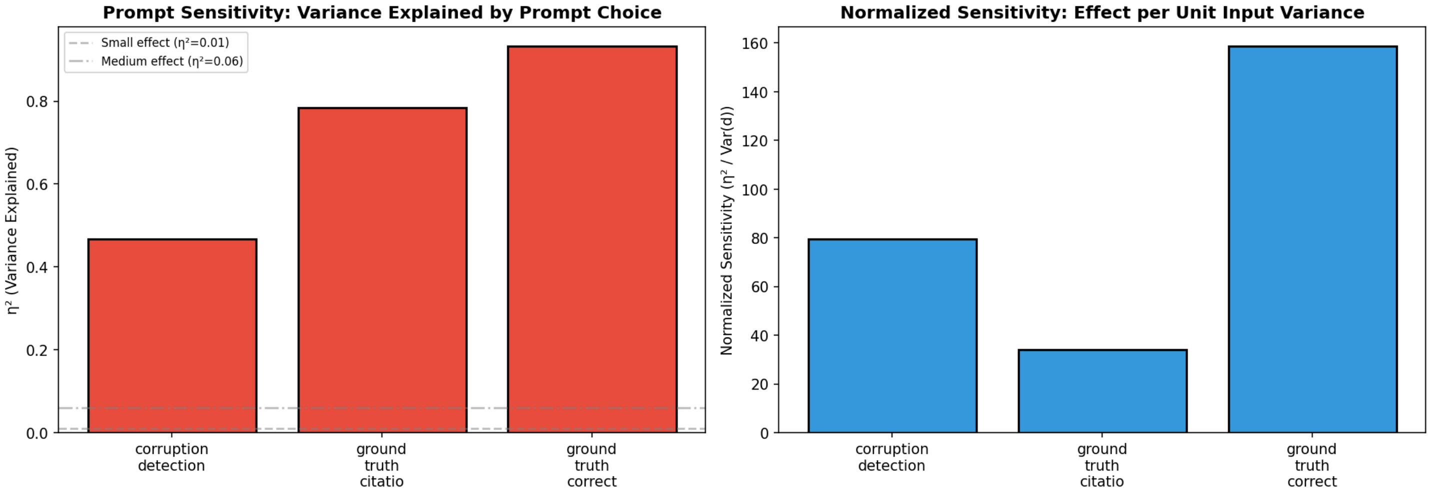


Figure 16: Aggregate Sobol sensitivity analysis. Shows first-order and total-order Sobol indices indicating the relative importance of each parameter.

7.2 Citation Detection Sobol Analysis

Sobol analysis for citation-based corruption detection.

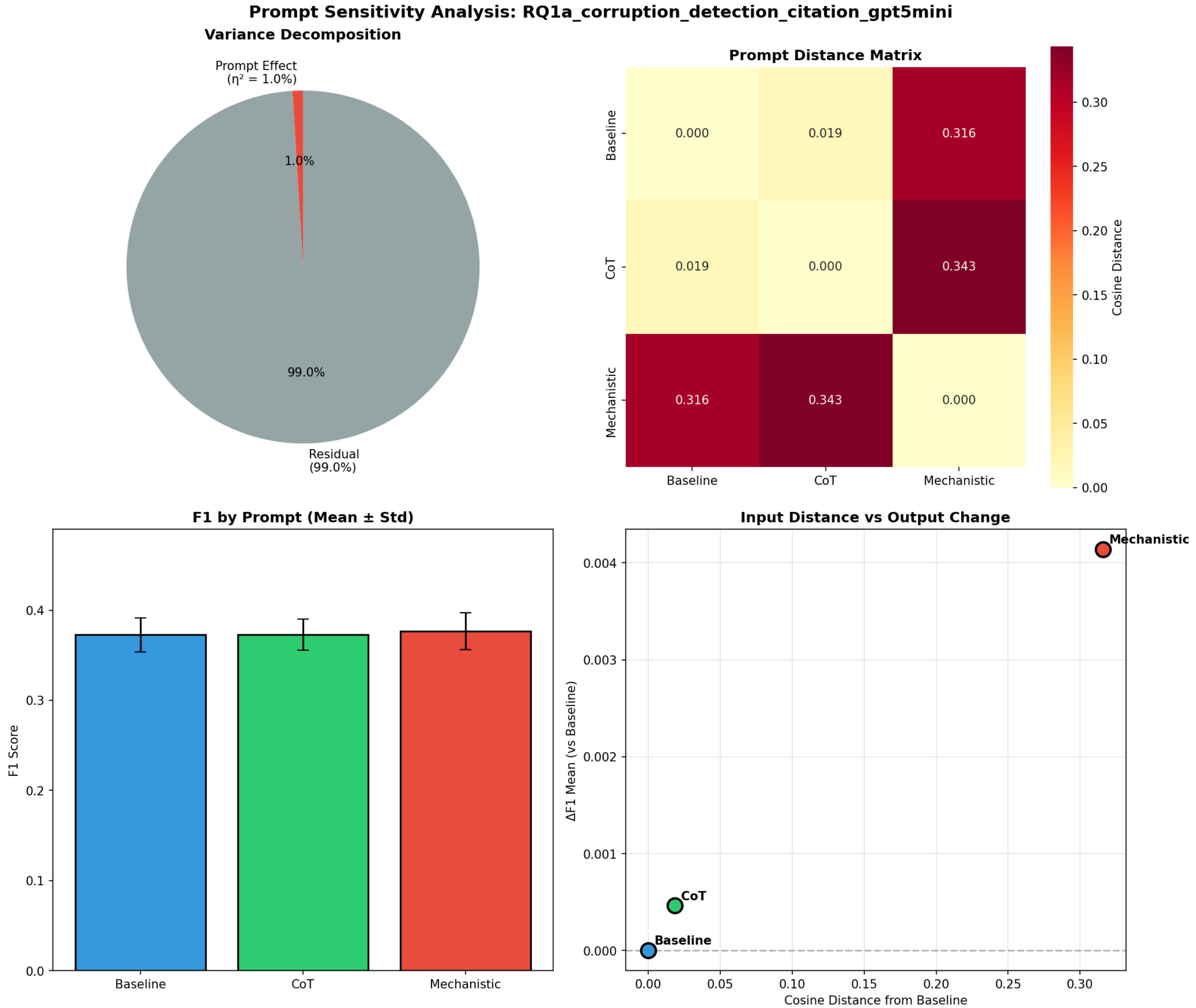


Figure 17: Sobol sensitivity analysis for citation-based corruption detection. Decomposes variance contributions from different experimental parameters.

7.3 Correctness Detection Sobol Analysis

Sobol analysis for correctness-based corruption detection.

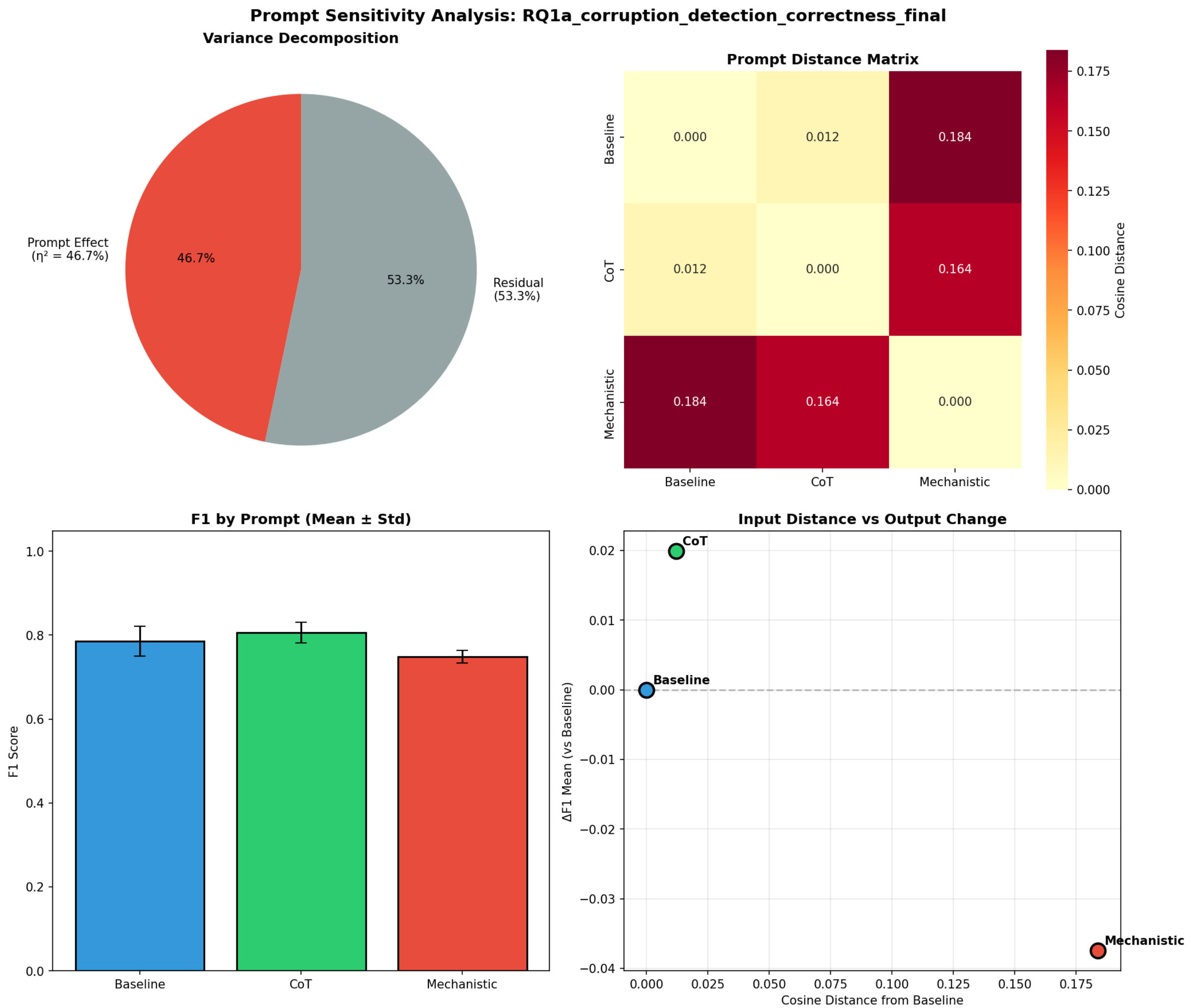


Figure 18: Sobol sensitivity analysis for correctness-based corruption detection. Quantifies parameter importance through variance decomposition.

7.4 Ground Truth Citation Sobol Analysis

Sobol analysis for ground truth citation validation experiments.

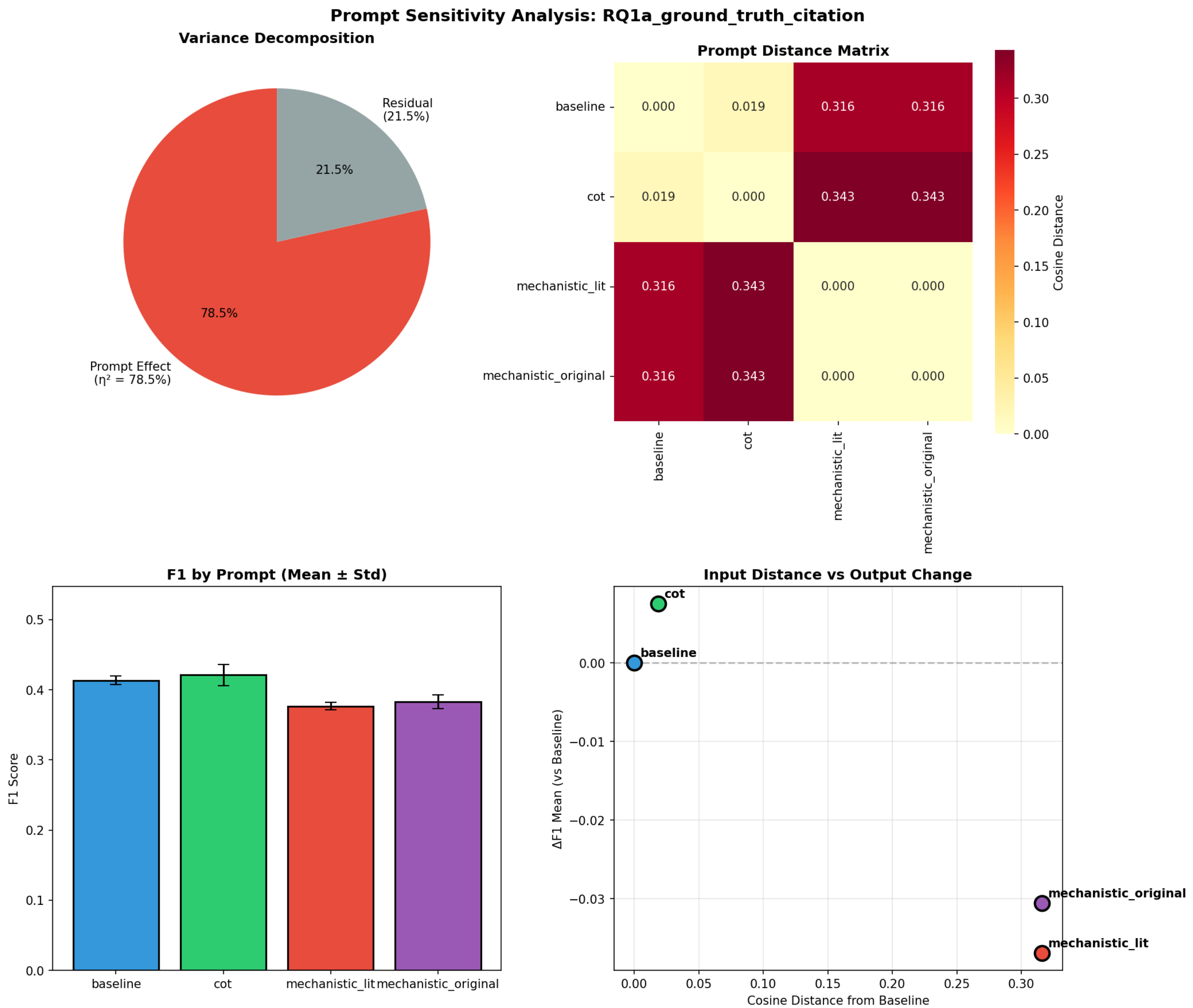


Figure 19: Sobol sensitivity analysis for ground truth citation validation. Establishes baseline variance decomposition.

7.5 Ground Truth Correctness Sobol Analysis

Sobol analysis for ground truth correctness validation experiments.

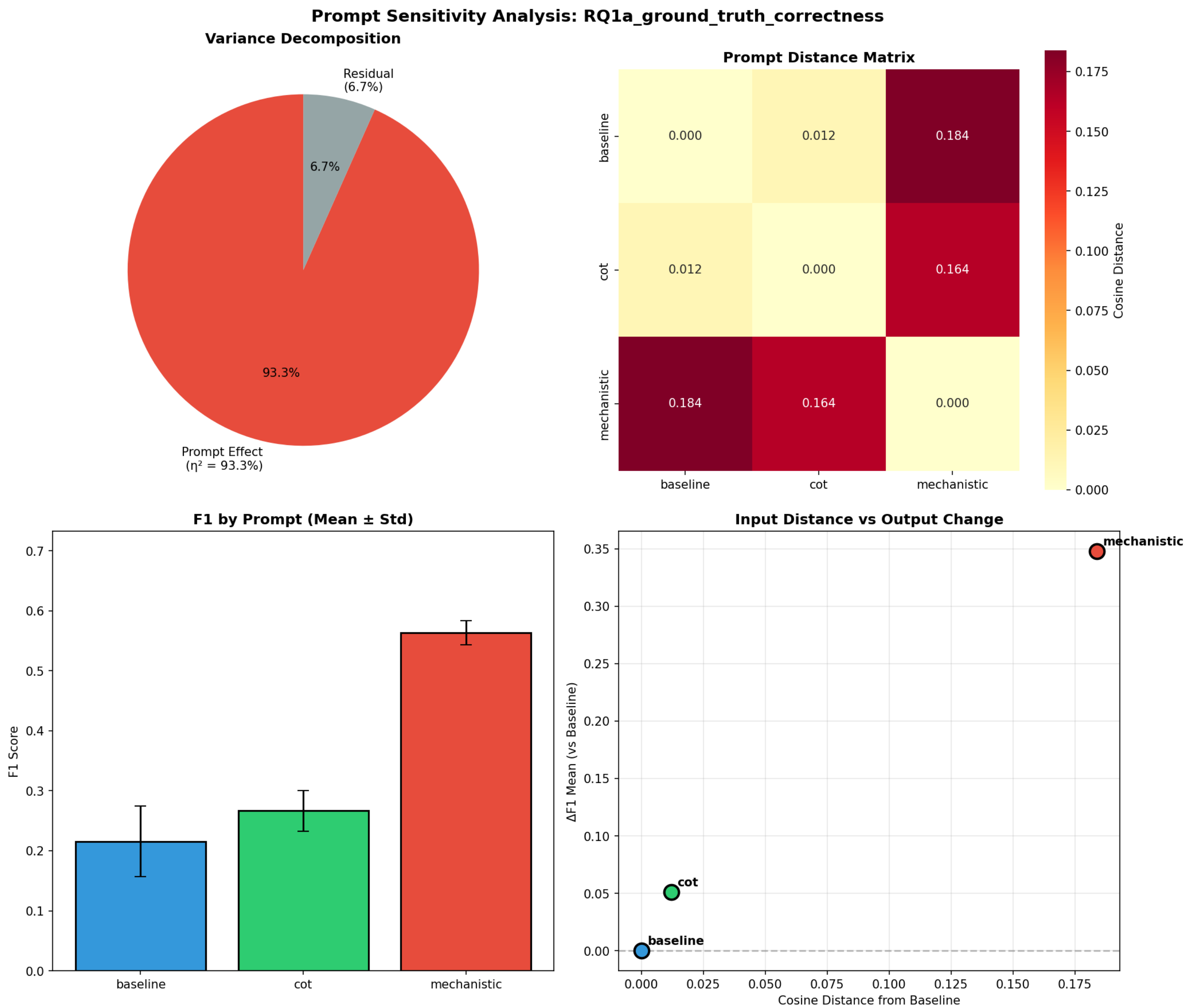


Figure 20: Sobol sensitivity analysis for ground truth correctness validation. Quantifies baseline parameter importance.

8 Summary

This document presents a comprehensive collection of all experimental results organized by research question. The figures demonstrate:

- **RQ1a Corruption Detection:** Performance using citation-based and correctness-based approaches with comprehensive 6-panel visualizations showing all key metrics
- **RQ1a Ground Truth:** Baseline validation performance on clean data without artificial corruption
- **RQ1b:** Effectiveness of the correction system in recovering from corrupted causal relationships
- **Sensitivity Analysis:** Robustness of the system to parameter variations using both standard and Sobol variance-based methods

All figures are saved at 300 DPI for publication quality with bold axis labels for clarity.